BOOK

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CHAPTER 1

CONTACT MAP PREDICTION BY MACHINE LEARNING

1.1 INTRODUCTION

Proteins fold into three-dimensional (3D) structures that encode their function. Genomics and more recently metagenomics [49] projects leave us with millions of protein sequences, of which only a small fraction have their 3D structure experimentally determined. There are several structural genomics projects attempting to bridge the huge gap between sequence and structure. The high throughput pipelines have to deal with important bottlenecks, e.g. a large number of sequences are unsuitable for structural determination with current methods [1]. Therefore, computational protein structure prediction remains an irreplaceable instrument for the exploration of sequence-structure-function relationships. This is especially important for analysis at the genome or inter-genome level, where informative structural models need to be generated for thousands of gene products (or a portion of them) in reasonable amounts of time.

The more reliable and accurate procedures for structure prediction are based on the transfer of knowledge between closely related proteins deposited in sequence and structure databases – the field known as template-based modelling. Those methods typically adopt heuristics based on sequence and/or structural similarity to model the unknown target structure based on known structures that are understood to be homologous to it. Automating the modelling process is difficult: there are several stages and critical points in the design (choice of templates, the creation of a correct sequence to structure alignment, etc.). For some of these points manual

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intervention leads to better predictions than fully automated methods [5]. The accuracy of template-based techniques strongly depends on the amount of detectable similarity, thus preventing the reliable application of these methods for a significant fraction of unannotated proteins. This is the realm of the so called *ab initio* or de novo protein structure prediction, where models are predicted without relying on similarity to proteins with known structure. *Ab initio* techniques are obviously not as accurate as those based on templates, but the design in this case is generally more straightforward. Moreover, improvements can be obtained by relying on fragment-based algorithms [40], that use fragments of proteins of known structure to reconstruct the complete structure of the target protein. A system for the *ab initio* prediction of protein structures is generally composed of two elements: an algorithm to search the space of possible protein configurations to minimise some cost function; the cost function itself, composed of various constraints being either derived from physical laws, experimental results, or being structural features (e.g. secondary structure or solvent accessibility) predicted by machine learning or other kinds of statistical systems [23].

1.1.1 Maps Definition and Description

A two-dimensional (2D) representation of a protein structure, or more in general of a 3D object, is a map of properties of pairs of its elements, for instance the set of distances among a protein's residues. Using 2D projections of 3D objects is an attractive way of encoding geometrical information of protein structures, as these can be made scale and rotation invariant and do not depend on the coordinate frame. Therefore, 2D projections can be modelled as the output variable of learning or statistical systems trained in a supervised fashion, i.e. using samples of (input, target) pairs collected from structure databases. A 2D encoding of a structure can be graphically represented as a 2D matrix. In the case of proteins, the geometrical relationship may involve fragments of the structure at different scales, using for instance amino acid [45] or secondary structure segment pairs [33], the former being much more commonly adopted than the latter.

Geometrical relationships between amino acids can be expressed as a set of distance restraints, e.g. in the form $L \leq d(i,j) \leq U$, where $d(i,j)$ is the distance between residues in positions i and j and L (resp. U) is the lower (resp. upper) bound on the distance. Restraints such as the above ones can be experimentally determined, for example from NMR experiments. Indeed, algorithms for modelling protein structures from distance restraints are borrowed from the NMR literature and use, for instance, stochastic optimisation methods [43, 11], distance geometry [3, 19], and variants of them [41, 7, 24].

A *distance matrix* or *distance map* consists of the set $\{d(i,j)\}_{i>j}$ of $N(N-1)/2$ distances between any two points in positions i and j of a protein with N amino acids. Note how the distance matrix corresponds to the above form of constraints with the lower distance bound equal to the upper one. Figure 1.1 shows a greyscale image of the distance matrix of the protein with PDB code 1ABV, where the distances are calculated between the C_α atoms.

There is a trade-off between the resolution of the input restraints, that is the uncertainty with which they specify the property of the pairs, and the ability of the reconstruction algorithm to recover the correct model from these inputs. In the best

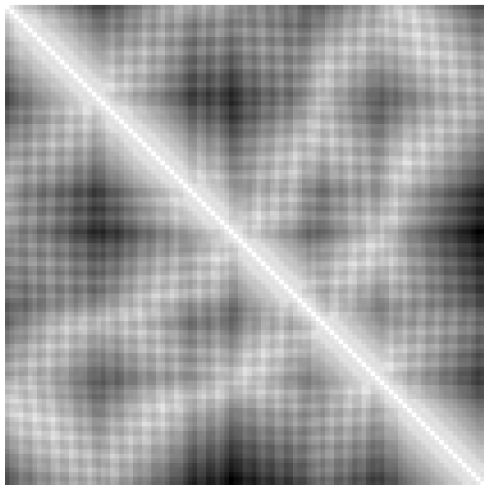

Figure 1.1 Distance matrix in greyscale image format. White is 0Å and black is the maximum distance in the protein.

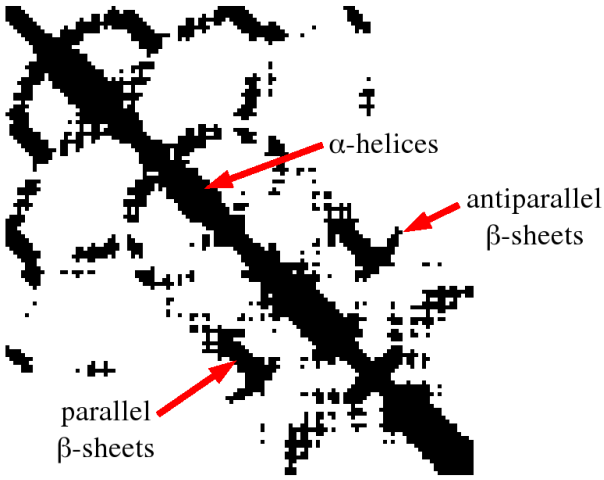

Figure 1.2 Different secondary structure elements like helices (thick bands along the main diagonal) and parallel − or anti-parallel − β-sheets (thin bands parallel − or anti-parallel − to the main diagonal) are easily detected from the contact map.

case, when the complete noise-free distance matrix is available, the optimisation problem can be solved analytically by finding a 3D embedding of the 2D restraints.

The distance matrix, or even detailed distance restraints, cannot be reliably determined by means of computational techniques unless experimental data is available or when there is strong homology to proteins of known structure. This is why in the past research has focused on predicting restraints derived from the distance matrix which are at the same time easier to learn than distances and are able to retain substantial structural information. The *contact map* of a protein is a labelled

matrix derived by thresholding the distance matrix and assigning labels to the resulting discrete intervals. The simplest alternative is the binary contact map, where one distance threshold t is chosen to describe pairs of residues that are in contact ($d(i,j) < t$) or not ($d(i,j) \geq t$). The binary contact map can also be seen as the adjacency matrix of the graph with N nodes corresponding to the residues. Binary contact maps are popular as noise-tolerant alternatives to the distance map, and algorithms exist to recover protein structures from these representations [43, 6, 42]. However, recovering high-quality models (1-2Å resolution) from the binary map alone, even that of the native fold, is often impossible [42], unless other sources of information are taken into account.

The definition of contact among amino acids is based on a single atom (normally C_α or C_β) and depends on a geometrical threshold. This is obviously a fairly crude approximation of physical interaction, for instance when the orientation of the side-chain is important. Nevertheless, it is possible to identify patterns in binary contact maps [18] – secondary structure elements can be recognised along the map diagonal as well as the interactions between them (Figure 1.2). Contact maps at 8Å have been assessed as a special category at CASP for several years [20].

1.1.2 Uses of Maps

It is often possible to reconstruct the 3D structure of a protein from a true contact map with a fair degree of accuracy. Reconstructed 3D structures can still be close to the native structure when a certain amount of noise is introduced into the contact map [42, 43] or even when predicted contact maps are used as constraints during the reconstruction process [6]. Contact maps have been used in many other applications. These include:

- Fold recognition: Cheng and Baldi use features of predicted contact maps (with thresholds at 8 and 12Å) as part of a set of 54 pairwise features measuring query-template similarity as input to support vector machines to determine if the query and template belong to the same fold or not [9].
- Model quality assessment: in [28] predicted contact maps are used to rank several different predicted 3D-models for a single target.
- Building non-aligned regions for template based models: the I-TASSER method [47, 50] for 3D model building uses templates when available, but unaligned loop regions are built by *ab initio* modelling using a potential function. The potential includes four components: (1) general knowledge-based statistics terms from the PDB (C_α/side-chain correlations, H-bond and hydrophobicity); (2) spatial restraints from threading templates; (3) sequence-based C_α contact predictions by SVMSEQ [48]; (4) distance and contact map from segmental threading. This method was ranked the best 3D server predictor at CASP6 and CASP7 [5].
- Protein structure comparisons: several methods have been developed which compare protein structures by making use of the fact that contact maps are rotation and translation invariant by searching for suboptimal solutions to the maximum contact map overlap problem [30, 31]. This reduces the search from 3D to 2D, making these methods fast and accurate ways of comparing structures.

- Protein folding simulation: as contact maps represent the 3D structure of a protein they have been used to simulate folding pathways – changing a few contacts in a map may represent substantial structural changes. An initial contact map is generated and successive changes to it reflect how the structure is modified during the folding processes. These changes to the map can be directed by different energy and/or map quality functions [39, 44].
- Predicting protein folding rates: in [37] folding rate of a protein is found to correlate with the number of predicted long-range contacts normalized by the square of the protein length.
- Predicting intrinsically disordered regions of proteins: in [38] contact map predictions by PROFcon [36] are combined with a pairwise potential to predict unstructured regions of proteins, with particular sucess for the case of long ones.
- Improving the prediction of protein secondary structure: high-quality predictions of contact maps and especially noise-free maps yield improved of near-perfect secondary structure predictions [8].
- Protein-protein and domain docking: while most of the methods used in docking are 3D coordinate based, in [25] protein domains are represented by their contact map and only contacts between the different domains have to be predicted. This method takes advantage of the invariance of rotation and translation of contact maps, avoiding the step of choosing the relative orientation and disposition of the domains with respect to each other.

1.1.3 Predicted Contact Map Quality Measures

In order to measure the quality of predicted contact maps several measures may be used. The most common ones take into account correctly or incorrectly classified contacts (true positives, TP; false positives, FP; true negatives, TN; and false negatives, FN), that are used to compute Recall (R, Equation 1.1), Precision (P, Equation 1.2) and their harmonic average, F measure (F, Equation 1.3), as follows:

$$R = \frac{TP}{(TP + FN)} \tag{1.1}$$

$$P = \frac{TP}{TP + FP} \tag{1.2}$$

$$F = \frac{2PR}{P + R} \tag{1.3}$$

Performance is usually computed for different sets of contacts based on the separation of two residues in the linear sequence. For instance at CASP [20] separations greater than $\{5, 11, 23\}$ have been used. Those contacts close to the diagonal of the map (i.e. for low sequence separations) reflect secondary structure/local interactions and are much easier to predict. Those further away from the map diagonal reflect long range interactions mainly between different secondary structure elements [16]. Usually, the greater the distance a map position is from the map diagonal the more difficult it is to predict as long-range contacts are less frequent, thus harder to characterise.

1.1.4 Contact Map Prediction

Although there has been improvement in contact map prediction accuracy over the last few years [20, 15], the problem is still largely unsolved. Among the reasons for this are the unbalanced nature of the problem (with far fewer examples of contacts than non-contacts, although this depends on the definition of contact) and, especially, the formidable challenge of capturing long-range interactions in the maps.

In order to mitigate the intrinsic difficulty of mapping 1D input sequences into 2D outputs, in virtually all existing predictive systems protein contact maps are inferred by modelling a set of independent tasks, each task being the prediction of whether two residues are in contact. For this task a variety of machine learning methods have been used: neural networks [12, 32, 36, 45]; genetic algorithims; self-organizing maps [27]; hidden Markov models [39]; support vector machines [10, 48, 51]; Bayesian inference [28]. The best performing map predictors tend to incorporate and combine information from different sources [20]. Common inputs are: conservation and co-frequency of residue pairs extracted from multiple sequences alignments (MSA); predicted secondary structure and solvent accessibility; predicted contact density and contact order.

In Section 1.2 we describe a system for binary contact map prediction based on recursive neural networks which we have developed. The system is capable of incorporating homology information from multiple templates from the Protein Data Bank, when available. Often predicted contact maps do not encode a physical 3D structure and some recent methods [14] try to eliminate non-physical contacts to improve the physicality of the predictions. An example of this type of strategy, which we have recently developed [26], is described in Section 1.4. In Section 1.5 we introduce a new definition of multi-thresholded maps along with methods we developed for their predictions. In Section 1.6 we discuss our participation to the CASP8 competition.

1.2 BINARY CONTACT MAP PREDICTION BY 2D-RECURSIVE NEURAL NETWORKS

We split the task of predicting binary contact maps into two stages: the prediction of contact density from the primary sequence; the reconstruction of the contact map from the predicted contact density and the primary sequence. We define contact density as the principal eigenvector (PE) of a protein's residue contact map at 8Å, multiplied by the principal eigenvalue. Predicting contact density from the primary sequence consists in mapping a vector into a vector. This task is less complex than mapping vectors directly into 2D matrices since the size of the problem is drastically reduced and so is the scale length of interactions that need to be learned.

Predicted contact density is incorporated into a system for contact map prediction [4, 32]. Our tests show that incorporating contact density yields sizeable gains in contact map prediction accuracy, and that these gains are especially significant for long-ranged contacts, which are known to be both harder to predict and critical for accurate 3D reconstruction.

1.2.1 Methods

We define two amino acids as being in contact if the distance between their C_α is less than a given threshold. For the definition of contact density we adopt a fixed 8Å threshold, while in the contact map prediction stage we test 8Å and 12Å thresholds (see [45] for a more detailed explanation of contact density prediction).

The data set used in the present simulations is extracted from the December 2003 25% pdb_select list[1]. We use the DSSP program [21] to assign relevant structural features and remove sequences for which DSSP does not produce an output. After processing by DSSP, the set contains 2171 protein and 344,653 amino acids. Multiple sequence alignments for the 2171 proteins are extracted from the NR database. The alignments are generated by three runs of PSI-BLAST [2] with parameters $b = 3000$, $e = 10^{-3}$ and $h = 10^{-10}$. For training and testing of the contact density predictor we split the data into a training set containing 1736 sequences and a test set of 435 (1/5 of the total). For training and testing of the contact map predictive system the sets are first processed to remove sequences longer than 200 amino acids (for computational reasons, as in [4]), leaving 1275 proteins in the training set and 327 proteins in the test set.

Predictive Architecture We build a system for the prediction of contact maps based on two-dimensional Recursive Neural Networks (2D-RNN), described in [4] and [32]. This is a family of adaptive models for mapping 2D matrices of variable size into matrices of the same size. 2D-RNN-based models were among the most successful contact map predictors at the CASP5 competition [29].

In the case of prediction of contact maps, the output O of the problem is the map itself, and the input I is a set of pairwise properties of residues in the protein. For a list of input properties we have considered, see below.

Let the element of indices j and k in the output be $o_{j,k}$ (in the case of contact maps this encodes whether residues in positions j and k in the sequence are within certain distance boudaries), in a standard 2D-RNN we postulate that $o_{j,k}$ is a function of four (vectorial) quantities, representing: the input element in the corresponding position $i_{j,k}$, e.g., in the case of contact maps, a vector encoding the identities of residues j and k, plus whatever individual or pairwise structural characteristics of these residues we decide to include; four contextual (vectorial) memories that encode information different parts of the input map, $h^{(n)}_{j,k}$ for $n = 1, \ldots, 4$. Each memory vector "specialises" on a different square of the input map, e.g. $h^{(1)}_{j,k}$ represents information about input elements $i_{v,t}$ for which $v \leq j$ and $t \leq k$ (the upper-left, or "north-western" context). We do not explicitly say what to represent in a memory vector, apart from defining its functional relationship to the other memory vectors, e.g. the upper-left memory $h^{(1)}_{j,k}$ has to be a function of $h^{(1)}_{j-1,k}$ (the upper-left memory in the same column and in the row above), of $h^{(1)}_{j,k-1}$ (the upper-left memory in the same row and in the column to the left), and of the input element $i_{j,k}$. The other three memory vectors in position j, k represent, respectively: the lower-left or "south-western" context ($h^{(2)}_{j,k}$); the lower-right or "south-eastern" context ($h^{(3)}_{j,k}$); the upper-right or "north-eastern" context ($h^{(4)}_{j,k}$). Overall, the set of functional relationships based on which we define a 2D-RNN is as follows:

[1] http://homepages.fh-giessen.de/~hg12640/pdbselect

$$
\begin{aligned}
o_{j,k} &= \mathcal{N}^{(O)}\left(i_{j,k}, h^{(1)}_{j,k}, h^{(2)}_{j,k}, h^{(3)}_{j,k}, h^{(4)}_{j,k}\right) \\
h^{(1)}_{j,k} &= \mathcal{N}^{(1)}\left(i_{j,k}, h^{(1)}_{j-1,k}, h^{(1)}_{j,k-1}\right) \\
h^{(2)}_{j,k} &= \mathcal{N}^{(2)}\left(i_{j,k}, h^{(2)}_{j+1,k}, h^{(2)}_{j,k-1}\right) \\
h^{(3)}_{j,k} &= \mathcal{N}^{(3)}\left(i_{j,k}, h^{(3)}_{j+1,k}, h^{(3)}_{j,k+1}\right) \\
h^{(4)}_{j,k} &= \mathcal{N}^{(4)}\left(i_{j,k}, h^{(4)}_{j-1,k}, h^{(4)}_{j,k+1}\right) \\
& \quad j,k = 1,\ldots,N
\end{aligned}
$$

Let us define, for simplicity, the set of these relationships as $O = \mathcal{N}(I)$.

We assume that the five functions (the output update $\mathcal{N}^{(O)}$ and the four lateral update functions $\mathcal{N}^{(n)}$ for $n = 1,\ldots,4$) are independent on the position on the map (the indices j,k of the input element they process), and represent them using five two-layered feed-forward neural networks, as in [4]. For an instance of the problem of $N \times N$ input elements (e.g. the contact map of a protein of length N) each neural network is replicated $N \times N$ times, resulting in a single neural network parametrising $\mathcal{N}$. This neural network may have a very large number of connnections (in the order of millions in the case of proteins) but the number of free parameters is greatly restricted (to the sum of free parameter numbers of the five networks) because each parameter is shared exactly $N \times N$ times. Given that the neural network representing $\mathcal{N}$ does not contain cycles, it can be trained by gradient descent using the standard backpropagation algorithm.

In the latest versions of our contact map predictors use 2D-RNNs with *shortcut connections*, i.e. where lateral memory connections span S-residue intervals, where $S > 1$. In this case the definition of a 2D-RNN changes into:

$$
\begin{aligned}
o_{j,k} &= \mathcal{N}^{(O)}\left(i_{j,k}, h^{(1)}_{j,k}, h^{(2)}_{j,k}, h^{(3)}_{j,k}, h^{(4)}_{j,k}\right) \\
h^{(1)}_{j,k} &= \mathcal{N}^{(1)}\left(i_{j,k}, h^{(1)}_{j-1,k}, \ldots, h^{(1)}_{j-S,k}, h^{(1)}_{j,k-1}, \ldots, h^{(1)}_{j,k-S}\right) \\
h^{(2)}_{j,k} &= \mathcal{N}^{(2)}\left(i_{j,k}, h^{(2)}_{j+1,k}, \ldots, h^{(2)}_{j+S,k}, h^{(2)}_{j,k-1}, \ldots, h^{(2)}_{j,k-S}\right) \\
h^{(3)}_{j,k} &= \mathcal{N}^{(3)}\left(i_{j,k}, h^{(3)}_{j+1,k}, \ldots, h^{(3)}_{j+S,k}, h^{(3)}_{j,k+1}, \ldots, h^{(3)}_{j,k+S}\right) \\
h^{(4)}_{j,k} &= \mathcal{N}^{(4)}\left(i_{j,k}, h^{(4)}_{j-1,k}, \ldots, h^{(4)}_{j-S,k}, h^{(4)}_{j,k+1}, \ldots, h^{(4)}_{j,k+S}\right) \\
& \quad j,k = 1,\ldots,N
\end{aligned}
$$

A graphical representation of a 2D-RNN with shortcuts is depicted in Figure 1.3.

In our tests the input $i_{j,k}$ contains amino acid information, secondary structure and solvent accessibility information, and contact density information for the amino acids in positions j and k in the sequence. Amino acid information is obtained from MSA of the protein sequence to its homologues to leverage evolutionary information. Amino acids are coded as letters out of an alphabet of 25. Beside the 20 standard amino acids, B (aspartic acid or asparagine), U (selenocysteine), X (unknown), Z

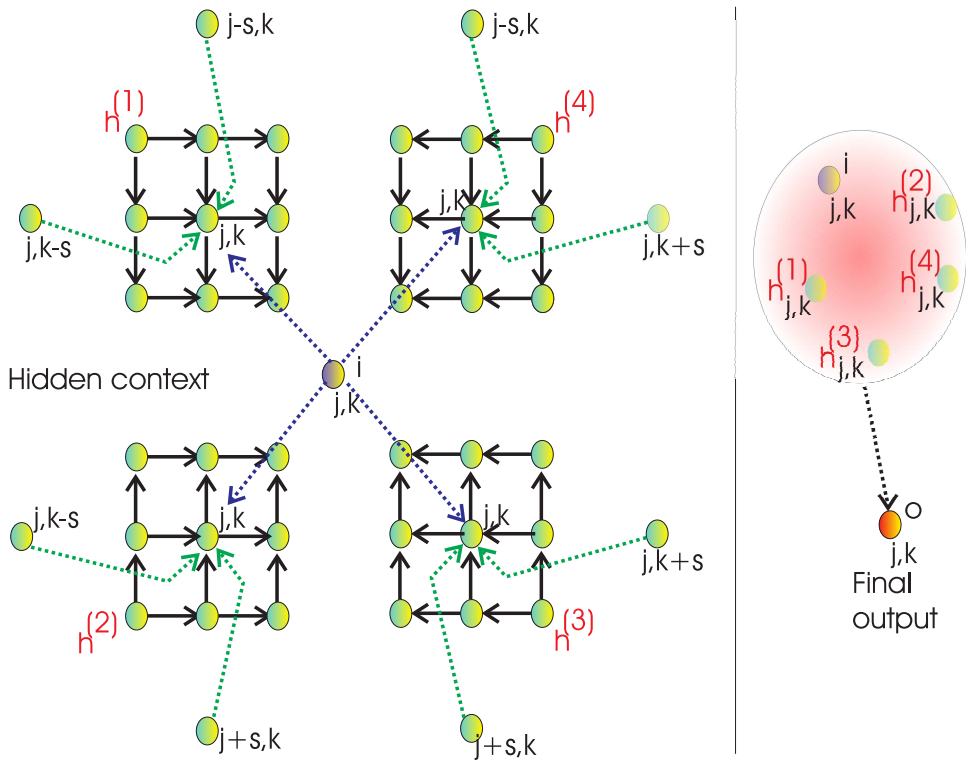

Figure 1.3 A graphical representation of a 2D-RNN with shortcut connections (see text for details). Nodes represent inputs element vectors ($i_{j,k}$), hidden memory vectors ($h^{(n)}_{a,b}$) and outputs ($i_{j,k}$). An edge between node A and B indicates that B is a function of A.

(glutamic acid or glutamine) and . (gap) are considered. The input presented to the networks is the frequency of each of the 24 non-gap symbols, plus the overall frequency of gaps in each column of the alignment, i.e. if n_{jk} is the total number of occurrences of symbol j in column k, and g_k the number of gaps in the same column, the jth input to the networks in position k is:

$$\frac{n_{jk}}{\sum_{v=1}^{24} n_{vk}} \tag{1.4}$$

for $j = 1 \ldots 24$, while the 25th input is:

$$\frac{g_k}{g_k + \sum_{v=1}^{24} n_{vk}} \tag{1.5}$$

This input coding scheme is richer than simple 20-letter schemes and has proven effective in [35].

The systems are trained by minimising the cross-entropy error between the output and target probability distributions, using gradient descent with no momentum term or weight decay. We use a hybrid between online and batch training, with 580 gradient updates (roughly one every 3 proteins) per training epoch. The training set is also shuffled at each epoch, so that the error does not decrease monotonically. When the error does not decrease for 50 consecutive epochs, the learning rate is divided by 2. Training stops after 1000 epochs. The gradient is computed using the back-propagation through structure (BPTS) algorithm [13].

1.2.2 Results and Discussion

Contact Map Prediction from Predicted Contact Density For comparison purposes, we encode each pair (i, j) of amino acids in the input by four different features: a 20×20 matrix corresponding to the probability distribution over all pairs of amino acids observed in the two corresponding columns of the alignment (MA); MA plus the actual discretised 4-class contact density component for both residue i and j (MA_CD); MA plus the actual secondary structure (3 classes) and binary thresholded (at 25%) relative solvent accessibility (MA_SS_ACC); and finally, the previous feature plus the actual 4-class contact density components (MA_SS_ACC_CD). We train eight 2D-RNN ensembles, with the same architecture, one for each input feature and contact threshold. Testing takes place by encoding each pair (i, j) on input with the *predicted* 4-class contact density. Secondary structure and solvent accessibility information input is exact both during training and testing.

	$\lvert i-j\rvert \geq 6$			$\lvert i-j\rvert \geq 12$			$\lvert i-j\rvert \geq 24$		
	P	R	F	P	R	F	P	R	F
MA	100	0	0	100	0	0	100	0	0
MA_CD	39.4	12.2	18.6	36.2	8.4	13.5	27.8	2.0	3.7
MA_SS_ACC	50.5	7.4	12.9	48.8	4.0	7.4	25.7	0.2	0.3
MA_SS_ACC_CD	43.3	11.3	17.9	38.9	7.2	12.1	25.5	2.2	4.1

Table 1.1 Performance results for contact map prediction by 2D-RNN. Contact threshold: 8Å.

	$\|i-j\| \geq 6$			$\|i-j\| \geq 12$			$\|i-j\| \geq 24$		
	P	R	F	P	R	F	P	R	F
MA	60.4	10.6	18.1	55.8	0.1	0.1	38.9	0.03	0.06
MA_PE	49.5	24.5	32.8	39.4	16.8	23.6	34.5	13.6	19.5
MA_SS_ACC	61.6	19.6	29.7	48.9	7.5	13.1	40.2	2.8	5.3
MA_SS_ACC_PE	54.2	23.5	32.8	42.2	14.6	21.7	36.7	10.9	16.8

Table 1.2 Performance results for contact map prediction by 2D-RNN. Contact threshold: 12Å.

Figure 1.4 Examples of contact map *ab initio* predictions at 12Å for protein 1A2P (108 amino acids). Exact map in the top-right half, predicted map in the bottom-left half. Prediction by MA_SS_ACC on the left, MA_SS_ACC_CD on the right (see text for details).

Tables 1.1 and 1.2 show performance indices for all eight networks. Indices considered are: precision (P); recall (R); their harmonic mean (F) (see Section 1.1.3). Performances are computed for three different sets of contacts, based on the separation of two residues in the linear sequence: $|i-j| \geq \{6, 12, 24\}$.

The introduction of contact density predictions increases the F measure in all cases. This is true for both 8 and 12Å maps, and for all separation thresholds. An improvement is observed both in the MA_CD vs. MA case and in the MA_SS_ACC_CD vs. MA_SS_ACC case. In all cases the introduction of the predicted CD yields larger performance gains than the exact secondary structure and solvent accessibility combined. Interestingly, the gains become more significant for longer range contacts: for $|i-j| \geq 24$. F grows from 0.2% to 9.9% and from 3.6% to 25% in the 8 and 12Å case respectively. CD-based networks are more confident away from the main diagonal (an example is shown in Figure 1.4), with a better balance between false positives and false negatives.

1.3 INCORPORATING TEMPLATE INFORMATION

Many protein sequences show detectable identity to sequences of known structure. If this happens, homology information, when available, is provided as a further input to our predictors, similarly to [34]. Almost any level of sequence identity to PDB templates (PSI-BLAST e-value as high as 10) yields more accurate maps than the *ab initio* predictor. Furthermore, in most cases the map predicted based on PDB templates is more accurate than the maps of the templates, suggesting that

the combination of sequence and template information is more informative than templates alone. All the results and comparisons shown bellow are for 12Å threshold maps. For each of the 1602 proteins in S1602 we search for structural templates in the PDB. We base our search on PDBFINDERII [22] as available on August 22 2005. An obvious problem arising is that all proteins in the S1602 set are expected to be in the PDB (barring name changes), hence every protein will have a perfect template. To avoid this, we exclude from PDBFINDERII every protein that appears in S1602. We also exclude all entries shorter than 10 residues, leading to a final 66,350 chains.

To generate the actual templates for a protein, we run two rounds of PSI-BLAST against a redundancy-reduced version of the NR database with parameters $b = 3000$ (maximum number of hits), $e = 10^{-3}$ (expectation of a random hit) and $h = 10^{-10}$ (expectation of a random hit for sequences used to generate the PSSM). We then run a third round of PSI-BLAST against the PDB using the PSSM generated in the first two rounds. In this third round we deliberately use a high expectation parameter ($e = 10$) to include hits that are beyond the usual comparative modelling scope ($e < 0.01$). We further remove from each set of hits thus found all those with sequence similarity exceeding 95% over the whole query, to exclude PDB resubmissions of the same structure at different resolution, other chains in N-mers and close homologues.

The distribution of sequence similarity of the best template and average template similarity is plotted in Figure 1.5. The average similarity for all PDB hits for each protein, not surprisingly, is generally low. This does not affect predictive performances.

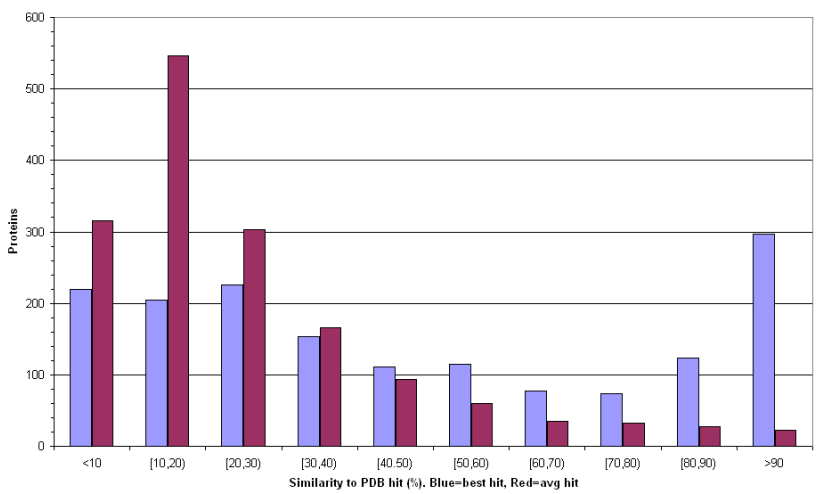

Figure 1.5 Distribution of best-hit (blue) and average (red) sequence similarity in the PSI-BLAST templates for the S2171 set. Hits above 95% sequence similarity excluded.

For a residue at position j a structural property d_j is input to the predictors as $d_j = \frac{\sum_{p=1}^{P} \omega_p d_{pj}}{\sum_{p=1}^{P} \omega_p}$ where: P is the total number of template residues aligned; d_{pj} represents the property d of the residue in the pth template, aligned against

SEQ_ID	<10	<20	<30	<40	<50	<60	<70	<80	<90	≥ 90	All
12_{AI}	85.9	87.5	86.8	85.6	87.2	86.5	86.2	86.1	86.4	87.3	86.8
12_{TE}	85.3	87.8	91.3	93.6	95.7	96.0	95.8	96.4	97.0	97.3	93.2

Table 1.3 Correctly predicted residues (%) for the *ab initio* (12_{AI}) and template based (12_{TE}) 12Å threshold contact map predictor as a function of percentage sequence identity (SEQ_ID) between the query sequences in S1602 and the best template found for each sequence.

position j in the query. $\omega_p = id_p^3/q_p$ is the weight associated to the p-th template, id_p is the identity between template p and the query protein, and q_p is the quality of template p measured as its resolution plus R-factor divided by 20.

Table 1.3 reports the comparison between *ab initio* and template based predictions of contact maps with a 12Å threshold. The only decrease in performance (by 0.6%) is when there is less than 10% sequence identity between the query sequence and the best template found for that sequence. The average increase in performance of template based predictions over *ab initio* predictions is 6.4%.

It should be noted that template generation is an independent module in the systems. We are currently investigating whether more subtle strategies for template recognition would still benefit contact map predictions, with or without retraining the systems on the new template distributions.

1.4 FILTERING CONTACT MAPS

We implement a filtering stage (NN-Filter) to correct non-physical contact predictions by training two different ensembles of artificial neural networks to process different regions of the contact map: the region close to the diagonal of the map, corresponding to small sequence separations; the region further away from the diagonal. We do so because the rules governing contact probability are likely to be different for positions near the main diagonal (mainly made by backbone atoms, reflecting secondary structure contacts) and for positions away from it (where contacts mainly occur between the side chains of amino acids placed in different secondary structure elements).

The input to the NN-Filter stage is a combination of local information about the predicted map such as estimated probabilities of contact for pairs of residues or secondary structure predictions for individual residues [35] and global information, such as total number of contacts predicted for a residue or pair of residues, and a protein's residue contact order.

1.4.1 Methods

The NN-Filter is trained and tested on the same datasets and MSA that were used to train and test the binary contact map predictor (which we will refer to as XXStout), as described above in Section 1.2.1. Therefore, any improvement by NN-Filter over XXStout cannot be attributed to differences in training set size or coverage. We create a further set ($TES481$) to assess the quality of filtered maps vs "raw" maps in an unbiased manner. From the July 2005 25% PDBSELECT list we remove all those sequences with sequence identity greater than 25% with

any of the sequences found the previously used datasets and those longer than 200 residues. This set contains 481 sequences (49, 159 amino acids).

Contact Maps As a cutoff between the filter for near-diagonal and long-range contacts we use the same distance thresholds between C_α (12Å and 8Å) as those defining binary contact maps that we are trying to filter. Predicted contacts are obtained from our binary contact map prediction system, XXStout, described in the previous section.

Artificial Neural Networks NN-Filter is composed of fully connected, feed forward neural networks trained via the backpropagation algorithm by minimising the cross-entropy error between the output and target probability distributions. Due to the large number of examples and the long time needed to train a single network on all training instances, the training set is further divided into 20 sets (away from diagonal) and 5 sets (beside diagonal), each containing approximately the same number of examples. A different network is trained on each of these subsets and then the networks are ensembled: 20 for map positions away from the diagonal; 5 for map positions beside the diagonal.

Input Representation

Local Inputs. The main input to the filtering neural networks is the "raw" map predicted by XXstout. We input a whole patch of 11x11 contact predictions. Thus a network that predicts the probability of contact between residues in positions i and j along the sequence is shown contact predictions (in the form of predicted probabilities of contact) for all pairs of amino acids (k, l) with $i - 5 \leq k \leq i + 5$ and $j - 5 \leq l \leq j + 5$, for a total of 121 inputs. We also input predicted secondary structure in the form of the predicted probabilities of being in one of the three standard secondary structure classes (Helix, Strand, Coil). Sequence and evolutionary information are provided to the network in the form of: identities of residues in positions $i-1$, i, $i+1$ and $j-1$, j, $j+1$; frequency profiles for residues in positions $i-1$, i, $i+1$ and $j-1$, j, $j+1$, extracted from MSA.

Global Inputs. The groups of inputs above provide local information about the map. We also adopt a number of input features that contain global information:

- Binary 8Å and 12Å map values for (i, j) obtained as follows: all the probabilities of contact, as predicted by XXStout, for pairs of amino acids at the same sequence separation ($|i - j|$) are added up yielding $\mathcal{L}$, then the L top-scoring pairs at the given separation are classified as contacts, where L is the integer part of $\mathcal{L}$. Notice that because of its definition, each element of these maps implicitly contains information about all those elements on the map with sequence separation $|i - j|$. This definition of contact is also used to determine the following features.
- Number of contacts per amino acid (NC) of amino acids i and j.
- Number of amino acids in contact with both i and j (NCIJ).
- Residue Contact Order (CO): for each amino acid i its CO (CO_i) is computed using Equation 1.6, where d_{ij} is the (euclidean) distance between amino acids

	XXStout			NN-Filter		
	F	P	R	F	P	R
$\|i-j\| \geq 6$	0.396	0.386	0.407	0.411	0.336	0.527
$\|i-j\| \geq 12$	0.323	0.313	0.334	0.353	0.278	0.484
$\|i-j\| \geq 24$	0.286	0.275	0.298	0.316	0.254	0.417
$\|i-j\| \leq 12$	0.858	0.857	0.858	0.876	0.875	0.876

Table 1.4 Performances for XXstout and NN-Filter on the $TES481$ data set for different sequence separations. 12Å maps.

	XXStout			NN-Filter		
	F	P	R	F	P	R
$\|i-j\| \geq 6$	0.213	0.354	0.153	0.271	0.277	0.265
$\|i-j\| \geq 12$	0.159	0.315	0.107	0.235	0.247	0.224
$\|i-j\| \geq 24$	0.097	0.230	0.057	0.168	0.219	0.137
$\|i-j\| \leq 8$	0.889	0.888	0.890	0.899	0.891	0.906

Table 1.5 Performances for XXstout and NN-Filter on the $TES481$ data set for different sequence separations. 8Å maps.

i and j, t is the contact threshold, L is the protein length and NC_i is the total number of contacts for i.

$$CO_i = \frac{\sum_{\forall j: d_{ij} < t} |i-j|}{L \cdot NC_i} \tag{1.6}$$

- Protein length.

1.4.2 Results and Discussion

We measure all performances on the $TES481$ data set. The F measure for NN-Filter improves over XXStout for both 12Å and 8Å maps, and for all bands of sequence separation considered. For long range contacts (sequence separation of 24 or greater) F grows from 0.286 for XXstout to 0.316 for NN-Filter for 12Å maps and from 0.097 to 0.168 for 8Å maps.

In all cases the improvements come from a small reduction in precision P, and a much greater increase in recall R. In the case of 8Å maps and contacts between residues with a sequence separation of 24 or greater, P decreases from 23% to 21.9%, while R more than doubles from 5.7% to 13.7%. Thus NN-Filter is able to predict more than twice as many long range contacts than XXstout, with only a minor loss of accuracy.

We tested both XXstout (as part of the Distill predictor) and NN-Filter at CASP7. As NN-Filter became available during the prediction season and after some of the targets has expired (i.e. no further predictions were accepted on them) we only submitted predictions for 63 of the 95 targets (68 domains). The results in Table 1.6 refer to these 63 targets. The results are for a contact threshold of 8Å between C_α. Both XXstout and NN-Filter are compared with other CASP7 predictors of contact maps. Proteins here are split in domains, as assessed by CASP. The table broadly confirms the results on $TES481$, with NN-Filter improving over

$\|i-j\|$	≥ 12			≥ 24		
Server	P	R	F	P	R	F
Gadjapairings	0.148	0.486	0.227	0.136	0.502	0.214
Betapro	0.269	0.150	0.193	0.230	0.120	0.158
SVMcon	0.401	0.124	0.189	0.330	0.104	0.158
NN-Filter	0.145	0.257	0.185	0.087	0.199	0.121
PROFcon	0.184	0.125	0.149	0.123	0.094	0.107
SAM_T06	0.382	0.077	0.128	0.353	0.073	0.121
Distill	0.088	0.269	0.132	0.054	0.189	0.084
Possum	0.086	0.180	0.117	0.058	0.164	0.085
GPCpred	0.064	0.119	0.083	0.034	0.131	0.054

Table 1.6 CASP7 results for XXstout (Distill) and NN-Filter vs. other CASP predictors. The results are macro-averages, i.e. averaged over all proteins in the set, and sorted by descending F. Results for 68 domains (63 targets) submitted by NN-Filter. Proteins are split into domains as assessed at CASP. All other predictors tested on the same 68 domains except: Gadjapairings (48 domains); PROFcon (66 domains).

"raw" XXstout predictions on all sequence separation bands.According to the CASP assessors, NN-Filter was the top predictor at one of the quality measures (X_d) [20].

Two examples of predictions with their filtered version are reported in Figures 1.6 and 1.7.

Figure 1.6 Example prediction for protein 1HRUA. Left side: bottom-left true map, top-right XXStout predicted 8Å map. Right side: bottom-left true map, top-right NN-Filtered 8Å map. In both cases the region beside the diagonal is predicted fairly well, while far from the diagonal, interactions between secondary structure elements are poorly predicted, especially interactions between strands that are not contiguous in the sequence.

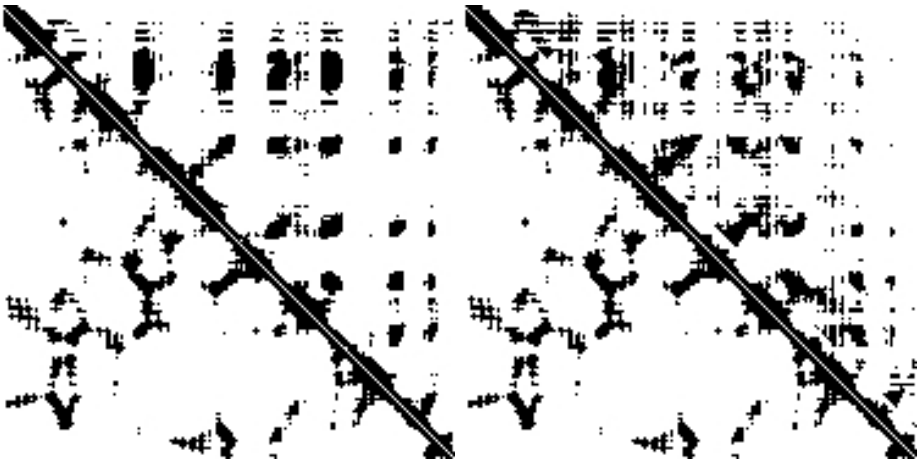

Figure 1.7 1HRUA. Left side: bottom-left true map, top-right XXStout predicted 12Å map. Right side: bottom-left true map, top-right NN-Filtered 12Å map. Simlarly to the 8Å case positions beside the diagonal are fairly well predicted, but interactions far from the diagonal are not. As for 8Å case there is a small improvement by NN-Filter near the diagonal. Far from the diagonal, the dense patches of contacts erronously predicted by XXStout have thinned out, although some isolated wrongly predicted contacts have appeared.

1.5 MULTI-CLASS DISTANCE MAPS

In this section we introduce a new class of distance restraints for protein structures: multi-class distance maps. We then build two predictors of 4-class maps based on recursive neural networks: one *ab initio*, or relying on the sequence and on evolutionary information in the form of MSA; one template-based, or in which homology information to known structures is provided as a further input.

Distance Matrix and Contact Map We focus on a representation of the distance matrix called multi-class contact map and based on a set of categorical attributes or classes. Each class corresponds to an interval of distances that a given pair of residues may fall into. Formally, given a set of distance thresholds $\{t_k\}_{k=0\ldots T}$ (where $t_0 = 0$ and $t_T = \infty$), a multi-class contact map of a protein with N amino acids is a symmetrix $N \times N$ matrix $\mathbf{C}$ where the element corresponding to the amino acids in positions i and j is defined as $\mathrm{C}_{ij} = k$ if $d(i,j) \in [t_k, t_{k+1})$. Obviously, this class of projections contains richer information than binary contact maps (so long as $T > 3$). Therefore, if thresholds are chosen carefully, native multi-class contact maps yield models of protein structures at significantly better resolutions than native binary maps [46]. Moreover, if a suitable set of distance thresholds is chosen, the number of instances in each class may be kept approximately balanced, which in turn may improve generalisation performances of learning algorithms over the (normally unbalanced) binary prediction case.

For our experiments, we derive a set of five distance thresholds to define multi-class contact maps based on four distance intervals. As shown in Figure 1.8, the four classes are empirically chosen from the distribution of distances among amino acids in the training set, ignoring trivial pairs $|i - j| \leq 3$ and by trying to keep informative distance constraints and the classes as balanced as possibile. The re-

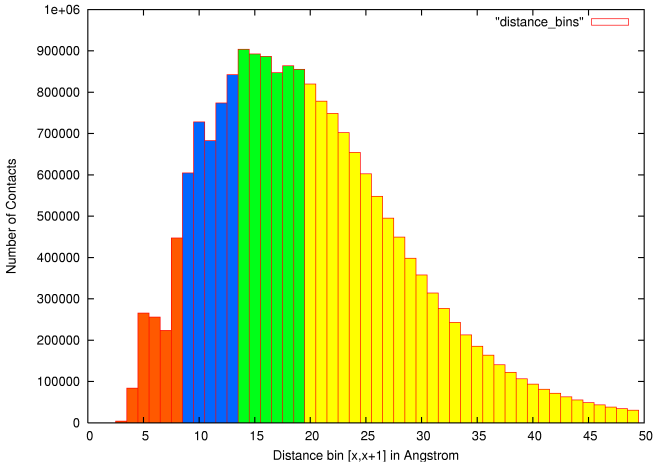

Figure 1.8 Distribution of contacts in [0,50] distance bins with trivial $|i-j| \leq 3$ residue contacts ignored. The classes were chosen in order to retain good distance constraints and balanced classes. Class 1 ([0,8)Å) corresponds, as a first approximation, to physical contacts.

sulting distance classes are: $[0,8)$, $[8,13)$, $[13,19)$ and $[19,\infty)$Å. The first class roughly corresponds to physical contacts, and matches the standard contact class adopted at CASP [20], the two middle classes are chosen to be roughly equally numerous and to span all categories of possible interaction (in [42] up to 18Å), while the fourth class represents non-contacts and is still the most numerous. Although this definition is somewhat arbitrary, our results were only minimally sensitive to small changes in the thresholds. A potential improvement beyond this choice is to automatically determine an optimal set of thresholds based on some criteria, e.g. the reconstruction ability on a set of benchmarking proteins.

1.5.1 Method

Predictive Architecture The predictive architecture for multi-class contact map prediction is essentially the same as that for binary class contact map prediction, that is we use ensembles of 2D-RNNs trained in 5-fold cross validation as previously described in Section 1.2.1.

Datasets and Template Generation The dataset used to train and test the predictors is extracted from the January 2007 25% pdb_select list [17]. The sequences are processed and MSA generated as described in Section 1.2.1 after which the set (S3129) contains 3129 proteins, 461,633 amino acids and just over 100 million residue pairs. Since training is computationally very demanding we create a reduced version of S3129 from which we exclude proteins longer than 200 residues. This set contains 2,452 proteins, and approximately 69 million residue pairs. All systems are trained in 5-fold cross-validation. This is obtained by splitting S3129 into 5 approximately equal folds, then (for training purposes) removing from the folds all proteins longer than 200 residues. Testing is on the full folds, i.e. including proteins longer than 200 residues. DSSP [21] assigned values for secondary structure and solvent accessibility are used during the training phase, however predicted values are used during

SEQ_ID	<10	<20	<30	<40	<50	<60	<70	<80	<90	≥ 90	All
M_{AI}	59.3	59.4	58.4	57.3	58.3	57.4	58.3	58.5	58.2	59.9	58.8
M_{TE}	60.2	64.2	75.9	82.5	87.8	88.8	88.1	89.7	91.5	92.1	80.8

Table 1.7 Correctly predicted residues (%) for the *ab initio* (M_{AI}) and template based multi-class predictor (M_{TE}) as a function of percentage sequence identity (SEQ_ID) between the query sequences in S3129 and the best template found for each sequence.

the testing phase. We then search the PDB for structural templates for each of the proteins as explained in Section 1.3 but against the PDB available on March 25th, 2008.

1.5.2 Results and Discussion

Table 1.3 reports the comparison between *ab initio* and template based predictions (12_{AI} vs. 12_{TE}) for binary contact maps with a 12Å threshold, as a function of sequence identity to the best PDB hit. The only decrease in performance for the template based predictor compared to the *ab initio* is in the [0,10)% identity range, where the accuracy slightly decreases by 0.6%. However, the same results for multi-class maps show that template based predictions are always more accurate on average than *ab initio* predictions (Table 1.7). Ultimately, templates improve multi-class predictions in all regions of sequence similarity (including [0,10)%), both for regions covered and regions not covered by templates.

Figures 1.9 and 1.10 show examples of a 12Å and a multi-class map predicted for a low best hit sequence identity of 22.7%.

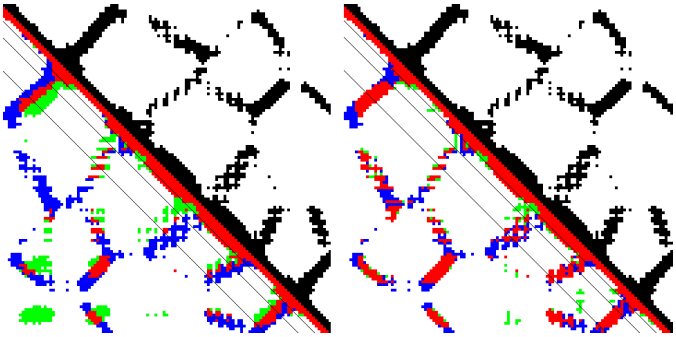

Figure 1.9 Protein 1B9LA 12Å contact maps for *ab initio* (left) and template-based (right) predictions. The best template sequence identity is 22.7%. The top right of each map is the true map and the bottom left is predicted. In the predicted half white and red are true negative and positive respectively, blue and green are false negative and positive respectively. The three black lines correspond to $|i - j| \geq 6, 12, 24$.

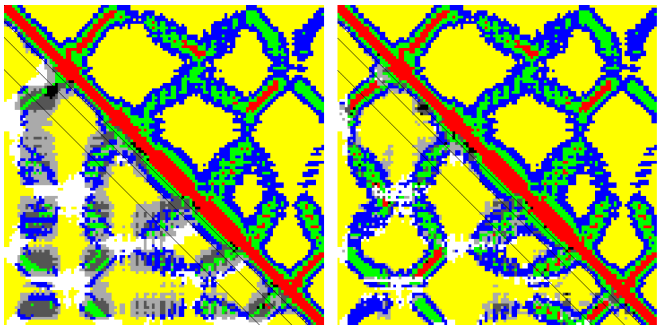

Figure 1.10 Protein 1B9LA Multi class contact maps for *ab initio* (left) and template-based (right) predictions. The best template sequence identity is 22.7%. The top right of each map is the true map and the bottom left is predicted. In the predicted half red, blue, green and yellow correspond to class 0, 1, 2 and 3 respectively. The greyscale in the predicted half corresponds to falsely predicted classes. The three black lines correspond to $|i-j| \geq 6, 12, 24$.

1.6 CASP8 EVALUATION

To compare our systems with other methods we participated in CASP8. We transformed multi-class maps into binary contact maps by simply recasting the second, third and fourth distance classes into non-contacts and keeping the first one ([0,8Å)) as contact. Table 1.8 shows the results of all the groups that participated in the map prediction category. When ranked by F measure, Distill is only outperformed by one group, SMEG-CCP. However, SMEG-CCP derives its maps by averaging 3D predictions produced by other groups and as such is not a direct predictor of contact maps. At CASP8 maps were ranked by Precision and only those maps with at least $\frac{L}{5}$ and $\frac{L}{10}$ were considered.

1.7 CONCLUSIONS

In this work we have described a machine learning pipeline for high-throughput prediction of protein contact maps. First, we introduced a system for binary map prediction and showed that the incorporation of 1D structural features, most notably contact density, into the prediction process improves prediction accuracy. Further, we described how to improve these maps by post-processing them by a filtering stage (NN-Filter) that uses first stage maps and long range information contained in them. Then, based on the observation that protein binary contact maps are somewhat lossy representations of the structure and yield only relatively low-resolution 3D models [6] we have introduced a new representation of the contact map, the multi-class contact map.

Moreover, extending on ideas we have developed for predictors of secondary structure and solvent accessibility [34] we have presented systems for the prediction of binary and multi-class maps that use structural templates from the PDB to yield far more accurate predictions than their *ab initio* counterparts. We have also shown that multi-class maps lead to a more balanced prediction problem than binary ones. Although it is unclear whether because of this, or because of the nature of

	$\|i-j\| \geq 6$			$\|i-j\| \geq 12$			$\|i-j\| \geq 6$		
	P	R	F	P	R	F	P	R	F
SMEG-CCP (121)	0.603	0.627	0.608	0.592	0.613	0.587	0.559	0.579	0.549
Distill (122)	0.438	0.465	0.446	0.419	0.446	0.427	0.386	0.422	0.396
LEE-SERVER (79)	0.513	0.369	0.396	0.509	0.357	0.382	0.480	0.332	0.353
LEE (121)	0.470	0.362	0.372	0.465	0.343	0.356	0.433	0.308	0.321
Pairings (112)	0.616	0.191	0.289	0.599	0.210	0.308	0.508	0.192	0.281
MULTICOM-CMFR (122)	0.167	0.356	0.216	0.153	0.266	0.174	0.110	0.205	0.129
MULTICOM-RANK (122)	0.129	0.475	0.198	0.116	0.399	0.174	0.095	0.311	0.139
RR_Fang_1 (122)	0.212	0.194	0.197	0.175	0.162	0.164	0.147	0.153	0.145
MUProt (122)	0.128	0.472	0.196	0.113	0.394	0.170	0.094	0.307	0.136
FLOUDAS (37)	0.277	0.162	0.193	0.244	0.132	0.158	0.176	0.097	0.112
RR_Fang_2 (122)	0.220	0.174	0.189	0.178	0.138	0.152	0.145	0.121	0.128
LCBContacts (121)	0.350	0.136	0.189	0.324	0.099	0.149	0.291	0.065	0.104
MULTICOM (121)	0.106	0.661	0.181	0.097	0.618	0.166	0.087	0.544	0.149
Infobiotics (119)	0.094	0.445	0.152	0.082	0.365	0.130	0.072	0.289	0.110
3Dpro (118)	0.082	0.429	0.136	0.077	0.344	0.123	0.064	0.264	0.099
SAM-T08-2stage (122)	0.075	0.481	0.129	0.074	0.501	0.128	0.070	0.408	0.118
AK_RF_2 (120)	0.084	0.265	0.125	0.074	0.245	0.111	0.067	0.228	0.100
SAM-T08-server (122)	0.071	0.462	0.123	0.070	0.499	0.121	0.066	0.444	0.113
SAM-T06-server (122)	0.065	0.419	0.111	0.062	0.454	0.109	0.056	0.419	0.098
Hamilton-Torda-Huber (122)	0.171	0.105	0.124	0.154	0.073	0.097	0.139	0.050	0.070
SVMSEQ (122)	0.037	0.794	0.071	0.035	0.770	0.067	0.037	0.717	0.070
SPINE-2DA-Zhou (121)	0.146	0.018	0.032	0.146	0.022	0.037	0.146	0.030	0.049

Table 1.8 CASP8 evaluation. Average values per target. True maps are computed from C_α instead of from C_β as in the CASP evaluation. The number in parenthesis after the predictor name is the number of targets submitted

the constraints encoded into them, template-based systems for the prediction of multi-class maps we tested are capable of exploiting both sequence and structure information even in cases of dubious homology, significantly improving over their *ab initio* counterpart well into and below the twilight zone of sequence identity. This turns out to be only partly true, at least in our tests, for binary contact map predictors.

It is important to note how the component for homology detection in this study is basic (PSI-BLAST), and entirely modular, in that it may be substituted by any other method that finds templates without substantially altering the pipeline. Whether more subtle homology detection or fold recognition components could be substituted to PSI-BLAST, with or without retraining the underlying machine learning systems, is the focus of our current studies. The overall pipeline, including the template-based component, is available at `http://distill.ucd.ie/distill/`. Protein structure predictions are based on multi-class maps and templates are automatically provided to the pipeline when available.

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Glossary

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